

Prescribing Mars PCM / MCD fluxes in KRC

User tutorial

Companion to *Prescribing Mars PCM / MCD atmospheric fluxes in KRC: design, database construction, coupling interface and validation*, to which the section numbers below refer.

1. Building the flux database

Before you start. The chain requires the DaVinci interface `Interface_KRC_MCD_PCM.dv`, the Python script `process_GCM_4_krc.py` with `numpy`, `xarray`, `scipy` and `h5netcdf`, a working KRC installation with its DaVinci interface, and the raw flux source described in Step 1. Steps 1 and 2 are performed once per simulation; Steps 3 to 5 are repeated for every location and dust scenario.

Step 1 – Obtain the flux source

- *PCM route (recommended).* Obtain the year-long concatenated PCM output with hourly visible, infrared and sensible surface fluxes. Contact E. Millour for these variables if a new simulation is required.
- *MCD route.* Use the pro interface of the MCD (<https://www-mars.lmd.jussieu.fr/mars/access.html>) to extract the same near-surface fluxes and the LTST on a longitude–latitude grid.
- The expected input is a NetCDF file with Time in sols and the local true solar time referenced to longitude 0.

Step 2 – Build the database file

```
python process_GCM_4_krc.py input.nc PCM_clim.hdf
python process_GCM_4_krc.py input.nc PCM_clim.hdf --griddeg # optional 1x1 deg regrid
```

- Name the output according to the scenario it represents (`PCM_clim.hdf`, `PCM_dusty.hdf`, `PCM_clear.hdf`, `PCM_MY26.hdf`): the interface selects the file from the MY argument and will not find a misnamed file.
- Place the file in `<path>/PCM/` or `<path>/MCD/`. The routine appends the model name to the path it is given, so the model name must be a directory.
- Check the printed diagnostics. The number of terminator points repaired in step 2 of the pipeline indicates the quality of the local-time reprojection; a large fraction indicates that the output frequency of the source simulation is too coarse.
- The resulting file must contain Time (8640 samples), `fvis`, `fir`, `sensible`, `tsurf`, `latitude` and `longitude`, as listed in Section 3.1 of the documentation.

2. Running KRC with prescribed fluxes

Step 3 – Build the KRC flux table

```
source('Interface_KRC_MCD_PCM.dv')

# path, model, MY scenario, latitude, longitude, include sensible heat flux
table = preprocess_atmosphericflux_from_MCD_PCM('/data/fluxes', 'PCM', 0, 4.5, 135.6, 1)
```

- MY: 0 for clim, 1 for dusty, 2 for clear, or 24 to 37 for a specific Martian year. With 'MCD', only 0 is currently available.
- Latitude in $[-90, 90]$, longitude in $[-180, 180]$. Out-of-range values return null with a message rather than failing silently.
- The routine performs a bilinear interpolation in latitude and longitude on the PCM grid, so the requested point need not coincide with a grid node.
- The last argument adds the sensible heat flux to the infrared column. Given the limitation discussed in Section 4.1 of the documentation, runs should be performed both with and without this contribution when it is significant.
- The returned table has seven rows: row 1 is the time axis, row 2 the visible flux (ASOL), row 4 the infrared flux (ATMRAD); the other rows are zero.

Step 4 – Run KRC with the prescribed fluxes

- Pass the table through the `UserPrescribed_Fluxes` argument of the KRC DaVinci interface.
- **Note.** Set the emissivity to 0.95, the value assumed by the PCM surface energy balance. A different value makes the prescribed infrared flux inconsistent with the surface receiving it.
- **Note.** Verify that the KRC seasonal step matches the $\Delta JD = 1.9083$ value hardcoded in `deltaJD_21s`. This is the failure mode encountered in one-point mode: the run completes without error while the fluxes are associated with incorrect seasons.
- Surface properties (albedo, thermal inertia) remain those of the KRC run. Only the atmospheric forcing is imposed.

Step 5 – Post-process and verify

```
out = Compute_VIS_IR_Tsurf_Table_Lsaxis_From_KRC_RUN(krc_out)
```

- Returns the surface temperature and the downwelling visible and infrared fluxes on the L_s axis.
- *First verification.* Compare the visible flux returned by KRC with `fvis` from the database at the same location. The two should coincide, as the solar flux is prescribed directly; a discrepancy indicates a seasonal misalignment rather than a physical effect.
- *Second verification.* Compare T_s with the `tsurf` field of the database, which is provided for this purpose. At latitudes below 45° the agreement should be within about 1 K (Section 5.2 of the documentation); a larger difference indicates a configuration problem, most probably in the emissivity or in the seasonal stepping.

3. Interpreting the results

The following points summarize what the coupling does and does not correct, and should be kept in mind when using the output quantitatively. Section numbers refer to the documentation.

- **Corrected.** The seasonality of atmospheric dust, which is now that of the PCM rather than the TAU2D climatology of KRC. This is the clearest improvement, most visible at mid latitudes for $L_s > 240^\circ$ (Section 7.3).
- **Not corrected.** The seasonal CO₂ caps and the northern perennial water-ice cap, which are governed by the surface parametrizations of KRC and not by the prescribed fluxes (Section 5.2). Results poleward of 45° should be treated with caution.
- **Degraded.** The equatorial region between $L_s = 90^\circ$ and 150° , where the PCM overestimates water-ice cloud opacity and therefore the downwelling infrared flux, leading to nighttime temperatures that are too warm (Section 7.3).
- **Residual inconsistency.** The prescribed infrared flux was computed for the surface temperature of the PCM. A remaining disagreement of 1 K between the KRC and PCM surface temperatures translates into a surface temperature error of 0.14 K to 0.26 K, depending on dust opacity (Section 6). This is not a limiting error source, but it scales with opacity and should be stratified accordingly.